◆ PayBD Full System Architecture

1) Main System Overview

PayBD will be a digital payment platform where users can:

- Send money
- Receive money
- Recharge mobile balance
- Pay bills
- View transaction history
- Check account balance

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2) Architecture

User App / Web
↓
Frontend (HTML / CSS / JS / App UI)
↓
API Layer
↓
Backend Server (PHP / Node.js / Laravel)
↓
Database (MySQL)
↓
Payment Engine
↓
Security Layer

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3) Frontend Part

User Interface Modules

- Login / Signup
- Dashboard
- Send Money
- Cash Out
- Payment
- Recharge
- History
- Profile

- HTML
- CSS
- JavaScript
- React
- Flutter (mobile app)

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4) Backend Part ⚙□

Main Backend Modules

Authentication System
Transaction System
Wallet Management
Notification System
Admin Control

Backend

- PHP
- Laravel

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5) Database Design

Core Tables	users table	transactions table	wallet table
users	id	id	id
transactions	name	sender_id	user_id
wallet	phone	receiver_id	balance
otp_codes	email	amount	last_update
admin	password	type	
notifications	balance	status	
	created_at	time	

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6) Payment Flow

User enters receiver number
↓
Enter amount
↓
PIN verify
↓
Backend checks balance
↓
Transaction save
↓
Balance update
↓
Success message

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7) Security Layer

- Password hash
- OTP verify
- Session protection
- CSRF protection
- SQL injection protection
- Transaction PIN

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8) Admin Panel

Admin will be able to:

- View all users
- Monitor transactions
- Block accounts
- Download reports

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9) Future Advanced Features 🚀

- QR Payment
- Merchant Payment
- Bank API connect
- NID verification
- Fraud detection
- AI spending analysis

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10) Real Production Architecture

App
↓
API Gateway
↓
Auth Server
↓
Transaction Server
↓
Wallet Server
↓
Database
↓
SMS Gateway

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11) Best Build Order (Step-by-Step) ✅

Step 1

Login + Register

Step 2

Wallet Balance

Step 3

Send Money

Step 4

History

Step 5

Admin Panel

Step 6

OTP

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